📘 MODULE 22: HEAVY FORWARDER

22.1 Concept

A **Heavy Forwarder (HF)** is a **full Splunk Enterprise instance** configured to forward data instead of (or as well as) indexing it. Because it is a full instance, it can **parse, filter, mask, and route** data before it reaches the indexer.

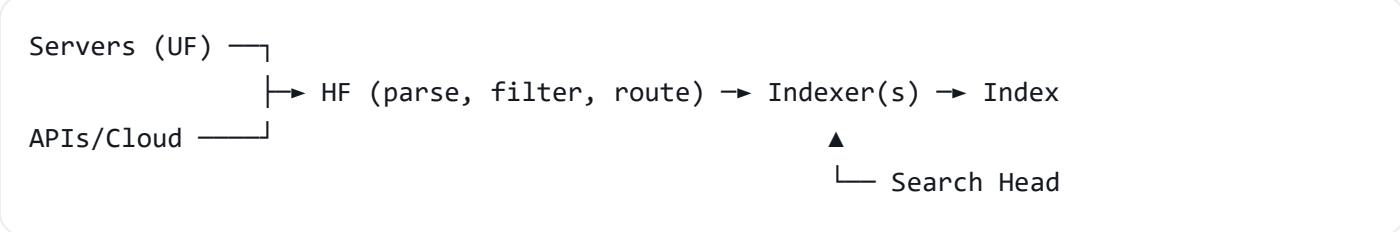
22.2 Why it matters

- Drop noisy events **before** they use license and storage.
- Mask sensitive values early.
- Route data to different indexes or different indexer groups.
- Run **add-ons that need a full instance** (API/cloud inputs, DB Connect).
- Act as an intermediary when UFs cannot reach the indexers directly.

22.3 UF vs HF

	UF	HF
Package	Separate lightweight package	Full Splunk Enterprise
Web UI	No	Yes
Parsing, filtering, routing	Minimal	Yes
Run Python add-ons / API inputs	No	Yes
Resource use	Very low	Significant
Licensing	Free forwarder license	Enterprise install; forwarding role is configured by you
Typical placement	On every endpoint	A few central servers

22.4 Architecture



Key rule: data parsed ("cooked") by an HF is **not re-parsed** by the indexer. So **parsing settings (`props` / `transforms`) belong on the HF** when an HF is in the path, not on the indexer behind it. This is a very common interview and troubleshooting point.

22.5 Set up an HF (outline)

1. Install **Splunk Enterprise** on a server.

2. Disable local indexing if you only want forwarding (optional): `[indexAndForward] index = false` in `outputs.conf`.
3. Enable receiving if UFs will send to it (port 9997).
4. Configure `outputs.conf` toward the indexers:

```ini
[tcpout]
defaultGroup = prod_indexers

[tcpout:prod_indexers]
server = idx1:9997, idx2:9997
```

5. Add `props.conf` and `transforms.conf` for filtering and routing (Module 23).

22.6 Filtering and routing examples

Drop debug lines (HF `props.conf` / `transforms.conf`):

```ini
# props.conf
[my_app_logs]
TRANSFORMS-drop_debug = drop_debug

# transforms.conf
[drop_debug]
REGEX = \sDEBUG\s
DEST_KEY = queue
FORMAT = nullQueue
```

Route to an index based on content:

```ini
# transforms.conf
[route_security]
REGEX = (failed|denied)
DEST_KEY = _MetaData:Index
FORMAT = security
```

Route to a different forwarding group:

```ini
```

```
[route_to_siem2]
REGEX = .
DEST_KEY = _TCP_ROUTING
FORMAT = siem2_group
```

(`siem2_group` must exist as a `[tcpout:siem2_group]` stanza.)

22.7 Troubleshooting

Symptom	Cause
Filtering/masking has no effect	Config placed on indexer, but the HF already parsed the data. Move it to the HF.
HF using lots of CPU/disk	It's indexing locally too, or too many inputs. Disable local indexing.
Data duplicated	HF indexes locally *and* forwards
Add-on input not running	Wrong instance type, credentials, or no outbound access

22.8 Interview Q&A

- **What is an HF and when do you use it?** A full Splunk instance that parses, filters, masks, and routes before forwarding. Use it for filtering/masking at the edge, API inputs, and intermediate forwarding.
- **Why not make every forwarder an HF?** Resource-heavy, harder to manage, and more to patch.
- **Where do you put props/transforms with an HF in the path?** On the HF, because cooked data is not re-parsed downstream.
- **Does an HF need a license?** It's a Splunk Enterprise instance, so licensing depends on setup. Check the current license docs for your deployment (common practice: point it to the license manager).

22.9 Revision

HF = full instance + forwarding · parse/filter/mask/route · cooked data isn't re-parsed · put parsing config on the first full instance in the path.